

Test Name - BOARD EXAM - MATH

Test Stream - PRE-NURTURE

Paper Code - 9610 WPF 40801023500M

Name - Aahan Badala

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1. (3) Rs. 500

2. (2) Rs 3600

3. (1) (2)

4. (3) $\begin{bmatrix} 3 & 7 \\ 1 & 7 \end{bmatrix}$

5. (4) 3:1

6. (2) (2, 3)

7. (3) zero

8. (1) 2

9. (4) 10

10. (2) y-axis

11. (1) (-3, -2)

12. (4) (3, -2)

13. (2) $P < 5$

14. (3) 7080

15. (4) 4:3:2

16)

(i)

$$x - 2 = 0$$

$$x = 2$$

$$ax^3 + 3x^2 - 13 = 2x^2 - 5x + 9$$

$$\Rightarrow 8a + 12$$

$$\Rightarrow a(2)^3 + 3(2)^2 - 13 = 2(2)^2 - 5(2) + 9$$

$$\Rightarrow 8a + 12 - 13 = 16 - 10 + 9$$

$$\Rightarrow 8a - 1 = 6 + 9$$

$$\Rightarrow 7a = 7$$

$$\Rightarrow \boxed{a = 1}$$

(ii) Let the numbers be $x-1$, x and $x+1$

$$\therefore x^2 = (x+1)^2 - (x-1)^2 + 60$$

$$x^2 = x^2 + 1 + 2x - x^2 - 1 + 2x + 60$$

$$x^2 = 4x + 60$$

$$x^2 - 4x - 60 = 0$$

$$x^2 - 10x + 6x - 60 = 0$$

$$x(x-10) + 6(x-10) = 0$$

$$(x+6)(x-10) = 0$$

$$x = 10, -6$$

x is a natural number so $x = 10$

$\therefore$ The three natural numbers are 9, 10, 11.

(iii) $y = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$

$$2x + y = \begin{bmatrix} 2 & 0 \\ -3 & 2 \end{bmatrix}$$

$$2x = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$2x - \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$$

$$2x = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

$$2x = \begin{bmatrix} 1+3 & 0+2 \\ -3+1 & 2+4 \end{bmatrix}$$

$$2x = \begin{bmatrix} 4 & 2 \\ -2 & 6 \end{bmatrix}$$

$$x = \begin{bmatrix} 4 \div 2 & 2 \div 2 \\ -2 \div 2 & 6 \div 2 \end{bmatrix}$$

$$\therefore x = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

17. i) let the two digit number be $10x + y$
 $\therefore xy = 18 \quad (1)$

$$10x + y - 63 = 10y + x$$

$$9x - 9y = 63$$

$$x - y = 7 \quad (2)$$

$$\therefore (7 + y) \times y = 18$$

$$= y^2 + 7y = 18$$

$$= y^2 + 7y - 18 = 0$$

$$= y^2 + 9y - 2y - 18 = 0$$

$$= y(y + 9) - 2(y + 9) = 0$$

$$= y = 2 \text{ or } -9$$

$$\therefore y = 2$$

$$x = 2 + y + 7$$

$$= 2 + 7$$

$$= 9$$

$\therefore$ The number is 92

ii) i) $2x^2 - 5x - x^2 - 2x + 3 = 0$

$$x^2 - 7x + 3 = 0$$

$\therefore$ Yes it is quadratic

ii) ~~$x^2 - 3x - \sqrt{x} + 4 = 0$~~

~~Since it has root~~

~~It is not quadratic~~

$$= x^2 - 3x + 4 = \sqrt{x}$$

$$= (x^2 - 3x + 4)^2 = (\sqrt{x})^2$$

$\therefore$ It is not quadratic

iii) $x(2x + 3) = x + 2$

$$2x^2 + 3x = x + 2$$

$$2x^2 + 2x - 2 = 0$$

Yes it is quadratic.

$$\textcircled{ii}) x + \frac{3}{x} = x^2$$

$$= \frac{x^2 + 3}{x} = x^2$$

$$= x^2 + 3 + x^3 = 0$$

∴ It is not quadratic

$$\textcircled{iii}) K+3 : K+2 :: 3K-7 : 2K-3$$

$$= \frac{K+3}{K+2} = \frac{3K-7}{2K-3}$$

$$(K+3)(2K-3) = (3K-7)(K+2)$$

$$= 2K^2 - 3K + 6K - 9 = 3K^2 + 6K - 7K - 14$$

$$= 2K^2 + 3K - 9 = 3K^2 - K - 14$$

$$= K^2 + 4K - 5 = 0$$

$$= K^2 - 5K + K - 5 = 0$$

$$= K(K-5) + 1(K-5) = 0$$

$$= K(K-5) + 1(K-5) = 0$$

$$= K = 5 \text{ or } -1$$

$$18. \textcircled{i}) \text{ Cost of articles} = ₹ 8,000$$

$$\text{GST rate} = 18\%$$

$$\text{GST to be paid} = 18\% \text{ of } 8000$$

$$= \frac{18}{100} \times 8000$$

$$= ₹ 1440$$

$$\therefore \text{Amount to be paid} = ₹ (8000 + 1440)$$

$$= ₹ 9440$$

ii) $(K-12)x^2 + 2(K-12)x + 2 = 0$ $D = 0$

~~For find~~ $D = (B^2 - 4AC)$

~~Let us take~~

Let us take

$$A = K - 12$$

$$B = 2(K - 12)$$

$$C = 2$$

$$[2(K-12)]^2 - 4 \times (K-12) \times 2 = 0$$

$$4(K-12)^2 - 8(K-12) = 0$$

$$= 4K^2 + 576 - 96K - 8K + 96 = 0$$

$$= 4K^2 - 104K + 672 = 0$$

$$= K^2 - 26K + 168 = 0$$

$$= K^2 - 12K - 14K + 168 = 0$$

$$= K(K-12) - 14(K-12) = 0$$

$$= K = 12 \text{ or } 14$$

~~But only~~

Using 12 :-

$$(12-12)x^2 + 2(12-12)x + 2 = 0$$

$$= 2 = 0$$

$\therefore 12$ will not work

Hence ~~the~~ K solution is equal to 14.

iii)

$$x^3 + (Kx + 8)x + K$$

$$x^2 + 1 = 0$$

$$x = -1$$

On substituting

$$(-1)^3 + (K \times (-1) + 8)(-1) + K$$

$$= 2K - 9 = R_1$$

$$\therefore x - 2 = x = 2$$

On substituting:

$$(2)^3 + (2K+2) + K$$

$$8 + 4K + 16 + K$$

$$= 5K + 24 - R_2$$

$$\text{Since } R_1 + R_2 = 1$$

$$2K - 9 + 5K + 24 = 1$$

$$7K = 10 - 24$$

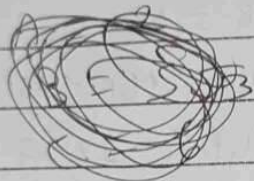
$$K = \frac{-14}{7}$$

$$K = -2$$

19.

i)

$$2x^2 + 5\sqrt{3}x + 6 = 0$$



$$2x^2 + 4\sqrt{3}x + \sqrt{3}x + 6 = 0$$

$$2x(x + 2\sqrt{3}) + \sqrt{3}(x + 2\sqrt{3}) = 0$$

$$= 2x + \sqrt{3} \quad \text{or} \quad x + 2\sqrt{3}$$

$$= x = -2\sqrt{3} \quad \text{or} \quad -\frac{\sqrt{3}}{2}$$

ii)

Let n th proportional be n .

iii)

$$2xy : x^2 :: y^2 : n$$

$$2xy \cdot n = x^2 \times y^2$$

$$n = \frac{x^2 y^2}{2xy}$$

$$n = \frac{xy}{2}$$

iv)

$$i) A = 1, B = 2, C = 4$$

$$D = b^2 - 4ac$$

$$\begin{aligned} D &= b^2 - 4ac \\ &= D = 4 - 4 \times 16 \\ &= D = -12 \end{aligned}$$

$$\begin{aligned} \textcircled{16} \text{ ii) } 10x^2 - 1 - 3x &= 0 \\ &= A = 10, B = -3, C = -1 \\ &= D = b^2 - 4ac \\ &= D = 9 + 40 \\ &= D = 49 \end{aligned}$$

$$\begin{aligned} \textcircled{17} \text{ iii) } x^2 - 5x + 1 &= 0 \\ &= A = 1, B = -5, C = 1 \\ &= D = b^2 - 4ac \\ &= D = 25 - 4 \\ &= D = 21 \end{aligned}$$

$$\begin{aligned} \textcircled{18} \text{ iv) } A &= 4, B = -1, C = -3 \\ &= D = b^2 - 4ac \\ &= D = 1 - 4 \times 4 \times -3 \\ &= D = 1 - (-48) \\ &= D = 49 \end{aligned}$$

20.

$$\textcircled{19} \text{ i) } \begin{bmatrix} -2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2x \end{bmatrix} + 3 \begin{bmatrix} -2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} y \\ \frac{y}{3} \end{bmatrix}$$

$$\begin{bmatrix} -2 \times -1 + 0 \times 2x \\ 3 \times -1 + 1 \times 2x \end{bmatrix} + \begin{bmatrix} -6 \\ 3 \end{bmatrix} = \begin{bmatrix} 2y \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 2x - 3 \end{bmatrix} + \begin{bmatrix} -6 \\ 3 \end{bmatrix} = \begin{bmatrix} 2y \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 2-6 \\ 2x-3+3 \end{bmatrix} = \begin{bmatrix} 2y \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} -4 \\ 2x \end{bmatrix} = \begin{bmatrix} 2y \\ 6 \end{bmatrix}$$

$$x = 3$$

$$y = -2$$

$$ii) \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 48$$

$$b = -13$$

$$c = -1$$

$$x = \frac{13 \pm \sqrt{169 + 4 \times 48}}{96}$$

$$x = \frac{13 + 19}{96}$$

$$x = \frac{32}{96} \text{ or } -\frac{6}{96}$$

$$x = \frac{1}{3} \text{ or } -\frac{1}{16}$$

$$iii) \quad Lx + my : Lz + mv :: Lx - my : Lz - mv$$

$$\frac{x}{y} = \frac{z}{v}$$

$$= \frac{Lx}{my} = \frac{Lz}{mv}$$

$$= \frac{Lx + my}{Lx - my} = \frac{Lz + mv}{Lz - mv}$$

$$= (Lx + my) \cdot (Lz + mv) \cdot (Lx - my) \cdot (Lz - mv)$$

Hence Proved.

21. i) Let the man be x years and son be y years

$$(x-2) = 3(y-2)^2 \quad \text{--- (1)}$$

$$x+3 = 4(y+3)$$

$$x = 4y+9 \quad \text{--- (2)}$$

$$4y+9-2 = 3(y^2+4-4y)$$

$$3y^2 - 16y + 5 = 0$$

$$3y^2 - 15y - y + 5 = 0$$

$$3y(y-5) - 1(y-5) = 0$$

$$(3y-1)(y-5) = 0$$

Since $\frac{1}{3}$ age is not possible

$$y = 5$$

Son's age = 5 years or age

$$\text{Father's age} = 4(5) + 9$$

$$= 29 \text{ years or age}$$

ii) $9x^2 - 6x + 2$

$$3x - 2 = 0$$

$$x = \frac{2}{3}$$

$$9\left(\frac{2}{3}\right)^2 - 6\left(\frac{2}{3}\right) + 2$$

$$9 \times \frac{4}{9} - 4 + 2$$

$$\text{Remainder} = 2$$

22.

i) $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^c = SA + TJ$$

$$\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1s & s \\ -s & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 9 & -1 & 3+2 \\ -3 & -2 & -1+4 \end{bmatrix} = \begin{bmatrix} 1s & s \\ -s & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 8-1s+7 & s+s \\ -s+s & 7+3-10 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

ii) Let the co-ordinates of third vertex be (x_3, y_3)

$$(0, -3) = \frac{-1+s+x_3}{3}, \frac{4+2+y_3}{3}$$

$$0 = \frac{4+x_3}{3}, -3 = \frac{6+y_3}{3}$$

$$x_3 = -4, y_3 = -15$$

The coordinates are

$$(-4, -15)$$

iii) $P = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$

$$P = \frac{2 \times -7 + 1 \times 2}{3}, \frac{2 \times 10 + 1 \times 1}{3}$$

$$P = (-4, 7)$$

$$O = (0, 0), P = (-4, 7)$$

$$d = \sqrt{(-4)^2 + (7)^2}$$

$$d = \sqrt{16 + 49}$$

$$d = \sqrt{65} \text{ units.}$$

23.

i) $P = [3 \times 3]$

ii) q

iii) $P_{13} = 4$

$$P_{32} = \frac{3}{5}$$

$$P_{22} = x$$

iv) Ratio = 1:1

$$\frac{-3+5}{2}, \frac{-4+1}{2}$$

$$= \frac{2}{2}, \frac{-3}{2}$$

$$C = \left(1, -\frac{3}{2}\right)$$

iii) Let $C = \begin{bmatrix} a \\ b \end{bmatrix}$

Then $AC = B$

$$\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3a+b \\ -a+2b \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}$$

$$3a + b = 7 \quad \text{--- ①}$$

$$-a + 2b = 0 \quad \text{--- ②}$$

$$\begin{aligned} 6a + 2b &= 14 \\ -a + 2b &= 0 \end{aligned}$$

$$7a = 14$$

$$a = 2$$

$$b = 1$$

$$\therefore C = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

24.

i) Let ratio be $K:1$

$$\frac{-2}{5} = \frac{2K-4}{K+1}$$

$$= -2K - 2 = 10K - 20$$

$$= \cancel{12K} 18 = 12K$$

$$\therefore K = \frac{18 \times 3}{12 \times 2}$$

$\therefore$ The ratio is $\frac{3}{2}$

ii) Let age of father be x and son be y .

$$\therefore x + y = 45$$

$$(x-5)(y-5) = 124$$

$$(40 - y - 5) \times (y - 5) = 124$$

$$= 40y - y^2 - 200 + 5y = 124$$

$$= y^2 - 45y + 324 = 0$$

$$= y^2 - 36y - 9y + 324 = 0$$

$$= y(y - 36) - 9(y - 36) = 0$$

$$(y - 36)(y - 9) = 0$$

$$y = 36 \text{ or } 9$$

Using 36

$$x + 36 = 45$$

$\therefore x = 9$ which is not possible

$$\therefore y = 9$$

$$x = 36$$

$\therefore$ The son is 9 years old while his father is 36 years old.

iii) ~~AB~~

$$AB = \begin{bmatrix} 1 & -2 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 2 - 8 & 3 - 10 \\ -8 + 4 & -12 + 10 \end{bmatrix}$$

$$\begin{bmatrix} -6 & -7 \\ -4 & -2 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 12 & -4 + 6 \\ 4 - 20 & -8 + 10 \end{bmatrix}$$

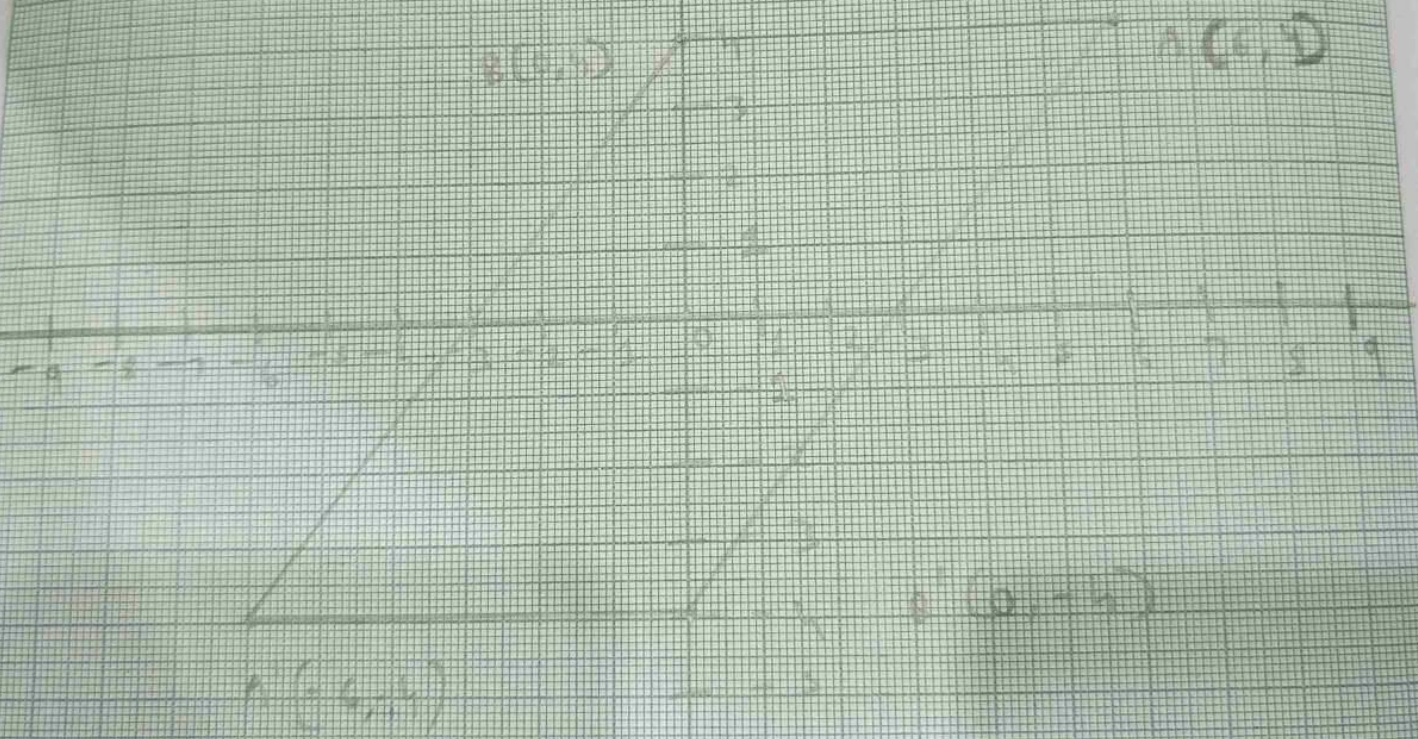
$$= \begin{bmatrix} -10 & 2 \\ -16 & 2 \end{bmatrix}$$

$$\text{Since } \begin{bmatrix} -6 & -7 \\ -4 & -2 \end{bmatrix} \neq \begin{bmatrix} -10 & 2 \\ -16 & 2 \end{bmatrix}$$

$$\therefore AB \neq BA$$

~~AB = BA~~

21 iii) i)



ii) $A' = (-6, -4)$, $B' = (0, -4)$

iii) It is a parallelogram.

iv) In $ABA'B'$, $BB' = 8$ units, $A'B' = 6$ units

$$\therefore BA' = \sqrt{6^2 + 8^2}$$

$$\therefore BA' = \sqrt{100}$$

$$BA' = 10 \text{ units.}$$

$$AB = A'B' = 6 \text{ units}$$

$$\therefore AB' = 10 \text{ units.}$$

$\therefore$ The perimeter of ~~ABA~~ $ABA'B'$ is:-

$$= 6 + 10 + 6 + 10$$

$$= 32 \text{ units.}$$